

The Volatility Complex

Topological Structure of Sector Stock Volatility
via Vietoris–Rips Complexes and Persistent Homology

Andrew Tawfeek

Erdős Institute — Quant Finance Boot Camp

Spring 2026

Outline

- 1 Motivation
- 2 Data & Preprocessing
- 3 Vietoris–Rips Complex
- 4 Rolling Complex Evolution
- 5 Persistent Homology
- 6 Results & Key Findings

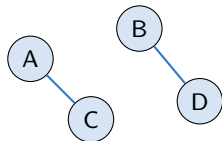
Why Topology for Volatility?

Observation: During market stress, asset volatilities become highly correlated — the market “moves as one.”

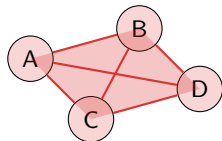
Goal: Capture this structural shift using *simplicial complexes* built from volatility correlations.

- Complexes encode **multi-body** relationships (not just pairwise correlations)
- Persistent homology reveals structure across **multiple scales**
- Time-varying complexes track **regime transitions**

Calm



Stress



Data Pipeline

Universe: ~50 S&P 500 Information Technology stocks, 2007–2024.

- 1 **Download** daily close prices via yfinance
- 2 **Log-returns:** $r_t = \ln(P_t/P_{t-1})$
- 3 **Rolling volatility** (30-day window, annualized):

$$\sigma_t^{(i)} = \text{std}(r_{t-29}^{(i)}, \dots, r_t^{(i)}) \times \sqrt{252}$$

- 4 **Rolling correlation** of volatilities (60-day window):

$$\rho_{ij}(t) = \text{corr}(\sigma_{t-59}^{(i)} : \sigma_t^{(i)}, \sigma_{t-59}^{(j)} : \sigma_t^{(j)})$$

- 5 **Distance metric:**

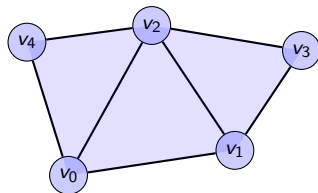
$$D_{ij}(t) = \sqrt{2(1 - \rho_{ij}(t))}$$

$D = 0$: perfectly correlated $\longleftrightarrow$ $D = \sqrt{2}$: uncorrelated $\longleftrightarrow$ $D = 2$: anti-correlated

Vietoris–Rips Construction

Given a distance matrix D and radius parameter $r > 0$:

- **0-simplices (vertices):** one per stock
- **1-simplices (edges):** (i, j) if $D_{ij} < r$
- **2-simplices (triangles):** (i, j, k) if all three pairwise distances $< r$ (flag complex)
- **k -simplices:** by the same clique rule



$$D_{01} < r, D_{02} < r, D_{12} < r \\ \Rightarrow \text{triangle } (v_0, v_1, v_2)$$

Radius $\longleftrightarrow$ correlation threshold:

$$r \approx 0.5 \quad \Rightarrow \quad \rho > 0.875 \quad (\text{very sparse})$$

$$r = 1.0 \quad \Rightarrow \quad \rho > 0.5 \quad (\text{moderate})$$

$$r \approx 1.414 \quad \Rightarrow \quad \rho > 0 \quad (\text{dense})$$

Tracking the f-Vector Over Time

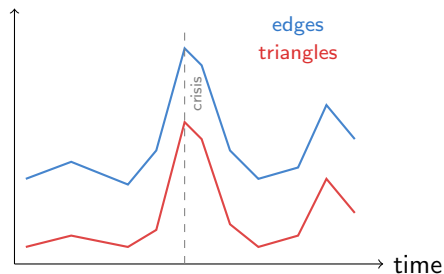
At each trading day t , we build a Rips complex from the rolling distance matrix and record the **f-vector**:

$$f(t) = \left(\underbrace{f_0}_{\text{vertices}}, \underbrace{f_1}_{\text{edges}}, \underbrace{f_2}_{\text{triangles}}, \underbrace{f_3}_{\text{tetrahedra}}, \dots \right)$$

Interpretation:

- Spike in edge count $\rightarrow$ volatilities synchronize (stress)
- Drop in edge count $\rightarrow$ idiosyncratic volatility (calm)
- Higher simplices (triangles, tetrahedra) capture *multi-body* co-movement beyond pairwise correlation

simplices



Schematic: simplex counts spike during market stress events.

Persistent Homology via Rips Filtration

For a **fixed time window**, vary the radius r from $0 \rightarrow 2$:

$$\text{VR}_{r_1}(D) \subseteq \text{VR}_{r_2}(D) \subseteq \cdots \subseteq \text{VR}_{r_k}(D) \quad (r_1 < r_2 < \cdots < r_k)$$

What we track:

- H_0 : **connected components**
Born: each stock starts isolated
Die: stocks merge as r grows
Long bars = well-separated clusters
- H_1 : **loops / cycles**
Born: cycle of edges without filling
Die: cycle gets filled by triangles
Long bars = robust cyclic structure
- H_2 : **voids** (enclosed cavities)

Orthogonality with rolling complexes:

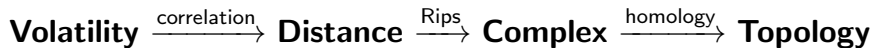
	Radius r	Time t
Rolling complex	fixed	varies
Persistence	varies	fixed

Together: a **2D picture** of how topology at each *scale* evolves over *time*.

Computed using **GUDHI** library
(Rips filtration $\rightarrow$ simplex tree $\rightarrow$ persistence).

Key Findings

- ① **Crisis detection:** Edge and triangle counts spike sharply during the 2008 financial crisis, 2020 COVID crash, and 2022 tech drawdown — the complex becomes nearly complete.
- ② **Cluster structure:** In calm periods, the 1-skeleton fragments into sub-clusters reflecting sub-sector groupings (semiconductors vs. software vs. hardware).
- ③ **Persistence signatures:** Long-lived H_0 bars during calm markets indicate well-separated volatility regimes. These bars collapse during stress as all stocks merge into a single connected component at lower radii.
- ④ **Higher-order structure matters:** Triangle and tetrahedron counts capture multi-body co-movement that pairwise correlation alone misses — the f-vector is a richer descriptor than edge count alone.



- Built Vietoris–Rips complexes from rolling volatility correlations of ~ 50 tech stocks over 17 years
- Tracked f-vector evolution to detect stress/calm regimes
- Computed persistent homology to reveal multi-scale structure
- The topological approach captures higher-order dependencies that pairwise methods miss

Code: `volatility_complex.py` | **Demo:** `notebooks/volatility_complex.ipynb`